

Zowe Capability Catalog – Design & Creation Documentation

1. Introduction

This document describes the design, rationale, and implementation of the Zowe Capability Catalog created as part of the Agentic Framework for IBM–Unisys Mainframe Autonomy project.

2. Purpose of the Catalog

The Zowe Capability Catalog serves as a structured knowledge base that maps Zowe CLI commands to their underlying IBM mainframe artifacts. It enables agent-based reasoning, safe execution planning, and governance-aware automation.

3. Command Selection Strategy

A representative subset of Zowe commands was selected to cover:

- Metadata discovery
- Data access
- Control and transaction operations
- Multiple IBM subsystems (DB2, CICS, IMS)

4. Schema Design

Each catalog entry captures:

- Zowe command
- Command category
- Command family
- IBM subsystem
- IBM artifact
- Operation type (READ / EXECUTE)

- Data returned
- Response format
- Intended agent
- Constraints and governance metadata

5. Mapping Zowe Commands to IBM Artifacts

Each command is manually mapped to the IBM artifact it interacts with, such as DB2 system catalogs, CICS program tables, or IMS transaction control blocks.

6. Catalog Persistence

The catalog is stored in an SQLite database for portability and simplicity. This allows consistent querying and reuse across backend services and agents.

7. API Exposure

A Flask REST API exposes the catalog via a GET endpoint, returning JSON-formatted catalog entries for frontend and agent consumption.

8. Frontend Visualization

A lightweight React frontend displays key catalog attributes, allowing users and reviewers to inspect command mappings and metadata.

9. Role in Agentic Framework

Agents use the catalog to:

- Discover safe commands
- Understand execution impact
- Reason about subsystems and artifacts
- Maintain explainability and auditability

10. Extensibility

The catalog design supports future extension to include additional Zowe plugins, job-based workflows, and automated enrichment.

11. Conclusion

The Zowe Capability Catalog transforms legacy mainframe operational knowledge into a machine-readable, agent-consumable format, enabling controlled and explainable autonomy.